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Discrete Mathematics

5Week [09 March - 15 March]

Quiz5

Started on	2024-03-19 14:27
State	Finished
Completed on	2024-03-21 18:33
Time taken	2 days 4 hours
Grade	4.00 out of 15.00 (27%)

Question 1

Correct

Mark 2.00 out of 2.00

What the following algorithm does?

```

procedure sum( $n$  : positive integer)
  if  $n < 10$  then return "count yourself"
  else
     $d := 0$ 
    for  $i := 1$  to  $n$ 
       $d := d + i$ 
    return  $d$ 
  
```

Select one or more:

- ☒ Returns sum of numbers from 1 to n if $n \geq 10$ ✓
- ☐ Returns list of numbers from 1 to n if $n \geq 10$
- ☐ Returns sum of numbers from 1 to n if $n < 10$
- ☐ Returns list of numbers from 1 to n if $n < 10$
- ☒ Returns "count yourself" if $n < 10$ ✓
- ☐ Returns "count yourself" if $n \geq 10$

Your answer is correct.

The correct answers are: Returns sum of numbers from 1 to n if $n \geq 10$, Returns "count yourself" if $n < 10$

Question 2
Partially correct
Mark 1.00 out of 2.00

It is given that

$$|f(x)| \leq 5|g(x)|$$

whenever $x > 3$. Choose statements, which are definitely true.

Select one or more:

☒ $f(x) = O(g(x))$ ✓

☒ $f(x) = O(g(x))$ with witnesses $C = 5$ and $k = 3$ ✓

☐ $f(x) = O(g(x))$ with witnesses $C = 6$ and $k = 4$

☐ $f(x) = O(g(x))$ with witnesses $C = 4$ and $k = 3$

☐ $f(x) = O(g(x))$ with witnesses $C = 5$ and $k = 2$

☐ $f(x) = O(2g(x))$ with witnesses $C = \frac{5}{2}$ and $k = 3$

Your answer is partially correct.

You have correctly selected 2.

The correct answers are: $f(x) = O(g(x))$.

$f(x) = O(g(x))$ with witnesses $C = 5$ and $k = 3$.

$f(x) = O(g(x))$ with witnesses $C = 6$ and $k = 4$.

$f(x) = O(2g(x))$ with witnesses $C = \frac{5}{2}$ and $k = 3$.

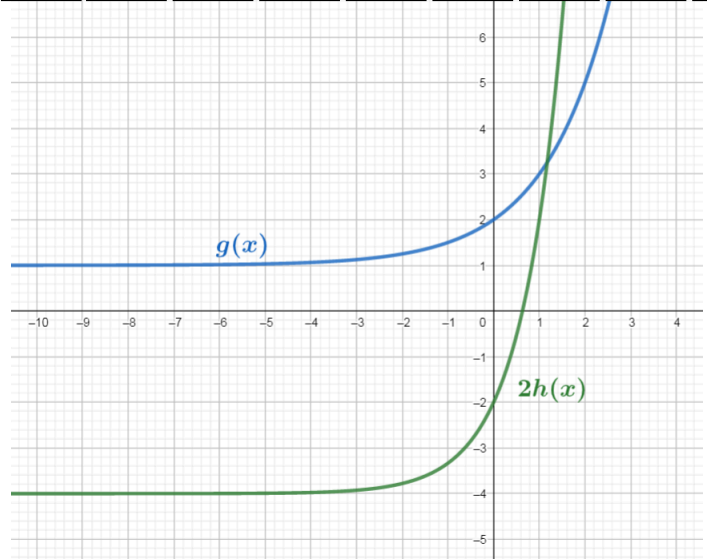
Question 3

Partially correct

Mark 1.00 out of 2.00

For two functions $g(x)$ and $h(x)$ determine which of the following relations are definitely correct (use graphs below).

1 log n n n log n n^3 n^{10} 3^n 4^n $n!$



Select one or more:

- ☐ $h(x) = O(g(x))$ with witnesses $C = 2$ and $k = 2$
- ☐ $h(x) = O(g(x))$ with witnesses $C = \frac{1}{2}$ and $k = 2$
- ☒ $g(x) = O(h(x))$ with witnesses $C = 2$ and $k = 2$
- ☐ $h(x) = O(g(x))$ with witnesses $C = 2$ and $k = 1$
- ☐ $g(x) = O(h(x))$ with witnesses $C = 2$ and $k = 1$
- ☐ $g(x) = O(4h(x))$ with witnesses $C = \frac{1}{2}$ and $k = 2$

Your answer is partially correct.

You have correctly selected 1.

The correct answers are: $g(x) = O(h(x))$ with witnesses

$C = 2$ and $k = 2$, $g(x) = O(4h(x))$ with witnesses

$C = \frac{1}{2}$ and $k = 2$

Question 4

Not answered

Marked out of 1.00

Drag and drop the following functions in such an order that each function in the list is a big-O of the next function in the list.

Your answer is incorrect.

The correct answer is:

Drag and drop the following functions in such an order that each function in the list is a big-O of the next function in the list.

[1], [log n], [n], [n log n], [n³], [n¹⁰], [3ⁿ], [4ⁿ], [n!]

Question 5

Not answered

Marked out of 1.00

Give a big-O estimate for the function

$$f(n) = (2^n + n^2)(n^3 + 3^n)$$

For the function g in your estimate $f(n) = O(g(n))$, choose a simple function g of the smallest order.

Select one:

- ☐ 6^n
- ☐ n^5
- ☐ $2^n n^3$
- ☐ $3^n n^2$
- ☐ $6^n n^5$

Your answer is incorrect.

The correct answer is: 6^n

Question 6

Not answered

Marked out of 3.00

Using only definition show that $3x^2 + 8x \log x$ is $\Theta(x^2)$.

Take a picture of your proof and attach it below.

Question **7**
Not answered
Marked out of
3.00

There is an algorithm for evaluating polynomials. It is called Horner's method.

This pseudocode shows how to use this method to find the value of

$$a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0 \text{ at } x = c.$$

procedure Horner($c, a_0, a_1, a_2, \dots, a_n$: real numbers)

$y := a_n$

for $i := 1$ **to** n

$y := y * c + a_{n-i}$

return y

Evaluate $3x^2 + x + 1$ at $x = 2$ by working through each step of the algorithm showing the values assigned at each assignment step.

Attach your solution below.

Question 8
Not answered
Marked out of
1.00

There is an algorithm for evaluating polynomials. It is called Horner's method.

This pseudocode shows how to use this method to find the value of

$$a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0 \text{ at } x = c.$$

procedure Horner($c, a_0, a_1, a_2, \dots, a_n$: real numbers)

$y := a_n$

for $i := 1$ **to** n

$y := y * c + a_{n-i}$

return y

Exactly how many multiplications and additions all together are used by this algorithm to evaluate a polynomial of degree n at $x = c$?

Select one:

- ☐ $2n$
- ☐ n
- ☐ n^2
- ☐ n^3
- ☐ 2^n

Your answer is incorrect.

The correct answer is: $2n$